Government of **Western Australia**
North Metropolitan Health Service
Mental Health Services

OFFICIAL

NMHS MHS PROCEDURE

Medication: Insulin Prescribing and Management

Scope (Audience)	All NMHS Mental Health Services (MHS) Clinical Staff
Scope (Area)	All NMHS MHS Inpatient Areas
Summary	This procedure provides NMHS MH clinical staff with the requirements for prescribing and management of insulin to ensure safe and effective treatment for patients of the service.
Risk Rating	High
NSQHS Standards	Clinical Governance (Std 1) Preventing and Controlling Healthcare Associated Infections (Std 3) Medication Safety (Std 4)
Procedure Sponsor	Director of Medical Services & Chief Pharmacist MHSDHS
Procedure Contact	Chief Pharmacist MHSDHS
Version	4.0
Review Date	20/03/2027

COMPLIANCE WITH THIS DOCUMENT IS MANDATORY

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1. Aim

This procedure provides NMHS MHS clinical staff with the requirements for prescribing and management of insulin to ensure safe and effective treatment for patients of the service.

2. Background [optional]

Medication errors involving insulin therapy can cause serious harm and may be fatal. It has been identified as a high risk medication by the Australian Commission on Safety and Quality in Health Care (ACSQHC).

Insulin is a high risk medication with the potential for serious adverse effects including hypoglycaemia, which may cause dizziness, sweating, hunger, faintness, headache, confusion, fast heartbeat, tremor and in severe cases, seizures, coma and death. This procedure is to be read in conjunction with the MP 0131/20 WA High Risk Medication Management Policy and NMHS MH Medication: Management of High Risk Medication Policy.

There are several different insulin preparations, which may cause confusion when prescribing or selecting the correct preparation for dispensing or administration.

Refer to the WA Statewide Adult Medicines Formulary (SMF) for current formulary/non-formulary status and/or restrictions of insulin preparations.

Administration of injections is recognised as a risk for occupational bodily fluid exposures and needle stick injuries to nursing staff. As a result, subcutaneous injections of insulin will only be administered using an insulin pen device in NMHS MHS.

3. Risk

Insulin is a **high risk** medicine which, if errors occur in preparation, dosing or therapy, can cause serious harm.

Non-compliance with this procedure will impact on the following risk categories:

Culture	☐	Organisational Development	☐
Emergency Management	☐	Policy & Legislative Compliance	☒
Fiscal Responsibility	☐	Quality of Patient Care	☒
Governance & Accountability	☒	Research & Innovation	☐
Information Communications Technology	☐	Staff Safety & Wellness	☒
Facility Management	☐	Workforce Planning	☐
Medical Equipment	☐	Other	☐

4. Procedure Principles

The insulin preparations available in Australia along with their storage details are detailed in **Appendix 1**.

4.1 Insulin Subcutaneous Order and Blood Glucose Record – Adult Chart (PMR60J)

All insulin orders at a NMHS MHS inpatient sites must be prescribed on the Insulin Subcutaneous Order and Blood Glucose Record – Adult PMR 60J.

MHS Education and Training Service provides education on the use of the insulin chart.

4.2 Prescribing insulin for subcutaneous insulin

All insulin must be prescribed with full dosage instructions. Care must be taken to ensure that the insulin type is fully documented, specifically:

- Full **brand/trade** name including concentration (e.g. Humulin 30/70® not just Humulin®, Humalog Mix 50® not just Humalog Mix®).
- Device (cartridge/disposable pen).
- Specify time of administration (i.e. immediately before meals or specific time to be given with respect to food).
- Dose - ensure that the word **'UNITS'** is written in full to avoid confusion.

4.3 Storage of insulin on wards

- Unopened cartridges/prefilled pens should be stored in a fridge at between 2 - 8 Degrees Celsius (°C). Do not place them in, or close to, the freezer compartment as they should not be frozen. Refer to **Appendix 1**.

- Insulin in use will be stored at room temperature (below 25°C) for up to 28 days before discarding.

EXCEPTION: Humulin® insulin cartridges (not vials) can only be stored at room temperature for up to 21 days before discarding.

- Do not expose insulin to direct sunlight or to high temperatures.

4.4 Labelling with patient and discard date

- When insulin on imprest is to be used for a patient, once removed from the refrigerator, the insulin must be **labelled** with a patient's addressograph sticker.
- The insulin cartridge/ prefilled pen **MUST** have the date opened, date removed from the fridge or discard date recorded on the device.

4.5 Administration of insulin by subcutaneous injection

- If a patient has their own non-disposable insulin pen (with cartridge refills) this may be used for administration whilst in hospital until pharmacy can provide hospital supply.
- If a patient does not have their own non-disposable insulin pen device, then a disposable pre-filled insulin pen will be used wherever possible.
- Ensure insulin is given subcutaneously at the prescribed dose using an insulin pen specific to the brand of insulin prescribed.

IMPORTANT: Refer to **Appendix 2** for Insulin Onset, Peak, Duration and Administration Times.

4.6 Management of Patients Admitted on Insulin Pump

Medical Officer Management

- The medical officer must assess and document in the integrated progress notes if the patient is deemed competent with no significant mental health risk to continue insulin pump therapy as an inpatient. If they are deemed capable and no significant mental health risk, the patient should be allowed to continue with insulin pump therapy.

- Assessment of competence will involve asking patient to describe or demonstrate the following:
  - Open the management menu of their pump
  - Access their basal rates/carbohydrate ratio/insulin sensitivity
  - Administer a bolus dose
  - Re-site their pump cannula
  - Technical competency regarding how they manage infusion line obstruction/site leaks
  - Adequate pump consumables
  - Low risk of harm to themselves or others with access to insulin
- If deemed competent and safe to self-manage, the medical officer will document the following:
  - Brand of Pump
  - Type of insulin
  - Basal rate of insulin
  - Carbohydrate ratio and insulin sensitivity settings
  - Insulin pump order is charted on the Insulin Subcutaneous Order and Blood Glucose Record – Adult PMR 60J denoting that continuous pump is running with units/hour as dose.
- If the patient is deemed incompetent or unsafe to self-manage the insulin pump, insulin pump treatment is to be discontinued and SCGH endocrinology team will be contacted for advice for further management.

IMPORTANT - Discontinuation of insulin pump for more than 30 minutes may result in significant hyperglycaemia, with the potential of diabetic ketoacidosis within 3 to 4 hours.

- Alternate insulin regimen is to be prescribed **PRIOR** to the disconnection of the insulin pump.

5. Monitoring

5.1 BGL Monitoring

The frequency of BGL monitoring depends on the patient's status, previous results and steps taken due to the results. Frequency and monitoring must be individualised. Patients with poorly controlled BGL will require more frequent monitoring than patients with a stable BGL.

The minimum BGL monitoring frequency for patients presenting with a diagnosis of diabetes will require an initial reading on admission and then standard BGL frequency as outlined below.

Standard BGL testing **for diabetic patients on insulin** is:

- Pre meals (breakfast, lunch and dinner) and 21:00 hours
- Testing frequency may need to be increased related to the patient's condition as indicated on the *Insulin Subcutaneous Order and Blood Glucose Record – Adult* ("Monitoring /Notification Instructions" section)

Standard BGL testing **for diabetic patients not on insulin**:

- Acute setting: pre meals and 21:00 hours

- Rehabilitation and Aged Care: four times a day for 48 hours, and then twice each day or twice a day biweekly if stable.

BGL results will be recorded in the *Insulin Subcutaneous Order and Blood Glucose Record – Adult.* Alert guidelines will be followed to notify a doctor as indicated.

Flash Glucose Monitoring (FGM) and Continuous Glucose Monitoring Systems (CGMS) are not approved for inpatient glucose monitoring due to the lag time behind blood glucose result. Patients admitted with FGM or CGMS will still require BGL reading from hospital supplied glucometer.

Steroids

Patients commencing, ceasing or changing their dose of oral corticosteroids require four times a day BGL monitoring for 48 hours. If the fasting BGL is >6mmol/L or a random BGL >10mmol/L, BGL monitoring will be continued, and a doctor notified to review the patient's insulin therapy.

(Reference: SCGOPHCG Diabetes Patient Management)

5.2 Ketone Monitoring

Blood ketone testing should be initiated for:

- Patients with newly diagnosed diabetes with BGL > 15mmol/L
- Patients with Type 1 Diabetes (on admission/presentation to ward/persistent hyperglycaemia)
- Patient receiving two or more insulin injections a day
- Patients on SGLT2i on admission, fasting or limited oral intake
- Patients with diabetes who have symptoms of DKA

6. Management of Hypoglycaemia or Hyperglycaemia

See the Guideline to Treating Hypoglycaemia or Guidelines for Treatment Review following Hyperglycaemia Alert pathways outlined in the Insulin Subcutaneous Order and Blood Glucose Record – Adult PMR 60J. for the management of hypoglycaemia or hyperglycaemia.

7. High Concentration Insulin

High concentration insulin is more concentrated than the standard insulin concentration of 100 units/mL. Fatal incidents and significant near miss events have been reported nationally and internationally involving incorrect formulation and dose.

To reduce the risk of selecting the wrong insulin, caution is needed when prescribing, dispensing and administering high concentration insulin.

- If the patient is admitted to hospital with high concentration insulin – it is preferable that the treating team continues the same preparation during their hospital admission.
- High concentration insulin will **NOT** be kept on imprest in NMHS-MHS inpatient wards.
- Pharmacy dispensary will dispense and label high concentration insulin for individual patients.
- High concentration insulin penfills **WILL NOT** be used to draw a dose into an insulin syringe for administration. All doses **MUST** be administered using the disposal injector pen

There are three high concentration insulin preparations available in Australia:

High Concentration Insulin	Concentration	Conversion to equivalent 100 units/mL Insulin
Insulin glargine (Toujeo Solostar®)	300 units/mL	Conversion to Optisulin® Toujeo® 100 units = Optisulin® 80 units A dose reduction of 15-25% is recommended when changing from Toujeo to Optisulin See: Medication safety Alert: High Concentration Insulin Toujeo®
Insulin lispro (Humalog U200 KwikPen®)	200 units/mL	No dose conversion required as the dose counter shows the number of units of strength, as does Humalog® 100units/mL
Neutral Human Insulin* **(Humulin R U-500 Kwikpen®)** ***only available via Special Access Scheme (SAS)**	500 units/mL	Dose equivalent on a unit-by-unit basis to Humulin® 100 units/mL.

8. Education

- Education on this procedure is provided at orientation to services for medical (prescribing) and nursing staff (administration including checking of order and storage).
- Education on this procedure is included in Professional Educational and Training (PEaT) for nursing staff.
- Education and monitoring of compliance with this procedure is included in Clinical teaching session and in Clinical Supervision for Medical and nursing staff.

9. Roles and Responsibilities

Roles	Responsibilities
All NMHS staff	Are responsible for: • Having knowledge of the procedure documents relevant to their roles and how to access these documents. • Complying with mandatory requirements outlined in these documents. • Assisting with procedure development/reviews wherever required. • Reporting to their immediate line manager any potential non-compliance or implementation issues.

10. Compliance, Monitoring and Evaluation

Compliance with this procedure is mandatory and will be monitored by NMHS MH Pharmacy, NMHS MH Infection Control and NMHS MH Occupational Safety & Health.

- Clinical incidents involving insulin will be reported on Datix CIMS.
- A Hazard Report will be completed for any needle stick injuries.

This procedure will be evaluated as required to determine its currency, relevance, effectiveness, efficiency, and sustainability. At a minimum, it will be reviewed every three years using a risk management approach in accordance with the NMHS Policy Governance Framework.

11. Definitions

Terms	Definitions
PBS	Pharmaceutical Benefits Scheme
SMF	WA Statewide Medicines Formulary
Penfill	a synonym of cartridge

12. Related Documents

Legislation	• *Health Services Act 2016*
WA Health mandatory policies	• WA Statewide Formulary One • MP 0131/20 WA High Risk Medication Management Policy • Statewide Medicines Formulary Policy (MP0077/18)
NMHS MHSDHS Policy documents	• MHSDHS IPC: Occupational Exposures Prevention and Management Policy
NMHS MH Policy documents	• MHS Medication: Management of High Risk Medication
Standards	• NSQHS Standard: Medication Safety
Other	• SCGOPHCG Diabetes Patient Management

13. Resources

- WATAG Advisory Note - Safety Alert: High Concentration Insulin
- AusDI
- Micromedex – Humulin R U-500 KwikPen Product Information
- Australian Medicines Handbook – comparative information of insulins
- National subcutaneous insulin medication charts (Australian Commission on Safety and Quality in Health Care)
- Diabetes Australia Position Statement – Use of Biosmiliar Insulins for Diabetes
- Diabetes yarning—all about diabetes
- Diabetes resources for Aboriginal and Torres Strait Islander people

14. Document Control

Document Version Control			
Date	**Version**	**Author**	**Notes**
30/12/2015	V1.0	Patrick Marwick, A/Executive Director	
12/2015	V1.0	NMHS MH Chief Pharmacist	Initial Endorsement
30/04/2019	V2.0	MHEG	Out of session
02/04/2019	V2.0	NMHS MH Chief Pharmacist	Scheduled Review: Updated template, references, NSQHS Standards V2.0, and hyperlinks. Addition of high strength insulin products. Updated Appendix 1: Trade and Generic names of Insulin available in Australia.
03/05/2019	V2.0		Endorsed via Submission Form
11/08/2022	V3.0	NMHS MH Chief Pharmacist	Scheduled review; updated template; Insulin chart implementation; Lantus®; Hypurin® Semglee® insulins discontinued. Information on blood glucose monitoring; Appendix 1. Updated.
18/03/2026	V4.0	NMHS MH A/Chief Pharmacist	Addition on management of insulin pump, review Appendix; changed to a procedure.

This document can be made available in alternative formats on request for a person with a disability.

Within the Department of Health WA, the term Aboriginal is used in preference to Aboriginal and Torres Strait Islander, in recognition that Aboriginal people are the original inhabitants of Western Australia. No disrespect is intended to our Torres Strait Islander colleagues and community. ISD: 5302.

Appendix 1. Trade/Generic names of human insulin available in Australia (current as at 10/07/2025)

TRADE NAME	GENERIC	VIAL	PENFILL	DISPOSABLE PEN	STORAGE
Before opening, insulin is stored in the fridge (between 2-8°C)					
Ultra-short acting					
Fiasp® *(**not interchangeable** with NovoRapid®)*	Aspart 100units/mL	Y	Y	Y Fiasp FlexTouch®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Humalog®	Lispro 100units/mL	Y	Y	Y Humalog KwikPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
NovoRapid®	Aspart 100units/mL	Y	Y	Y NovoRapid FlexPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Apidra®	Glulisine 100units/mL	Y	Y	Y Apidra SoloStar®	Discard 28 days after 1st use or removal from fridge (< 25°C)
High Concentration – Ultra Short-Acting					
Humalog U200®	Lispro **200units/mL**	N	N	Y Humalog U200 KwikPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Short-Acting					
Actrapid®	Neutral Human 100units/mL	Y	Y	N	Discard 28 days after 1st use or removal from fridge (< 25°C)
Humulin R®		Y	Y	N	Discard **cartridges** 21 days after 1st use (**vials** after 28 days) or removal from fridge (< 30°C)
High Concentration – Short Acting					
Humulin R U-500® (only available via SAS)	Neutral Human **500units/mL**	Y	N	Y Humulin R U-500 KwikPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Long-Acting					
Humulin NPH®	Isophane Human 100units/mL	Y	Y	N	Discard **cartridges** 21 days after 1st use (**vials** after 28 days) or removal from fridge (< 30°C)
Protaphane®		Y	Y	Y Protaphane InnoLet®	Discard 28 days after 1st use or removal from fridge (< 25°C)
Levemir®	Detemir 100units/mL	N	Y	Y Levemir FlexPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Optisulin®	Glargine 100units/mL	N	Y	Y Optisulin SoloStar®	Discard 28 days after 1st use or removal from fridge (< 30°C)

High Concentration – Long-Acting					
Toujeo®	Glargine **300units/mL**	N	N	Y Toujeo SoloStar®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Mixed					
Humulin 30/70®	Neutral Human 30% + Isophane Human 70% 100units/mL	Y	Y	N	Discard **cartridges** 21 days after 1st use (**vials** after 28 days) or removal from fridge (< 30°C)
Mixtard 30/70®		N	Y	Y Mixtard 30/70 InnoLet®	Discard 28 days after 1st use or removal from fridge (< 25°C)
Mixtard 50/50®	Neutral Human 50% + Isophane Human 50% 100units/mL	N	Y	N	Discard 28 days after 1st use or removal from fridge (< 25°C)
Humalog Mix25®	Lispro 25% + Lispro Protamine 75% 100units/mL	N	Y	Y Humalog Mix25 KwikPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Humalog Mix50®	Lispro 50% + Lispro Protamine 50% 100units/mL	N	Y	Y Humalog Mix50 KwikPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
NovoMix 30®	Aspart 30% + Aspart Protamine 70% 100units/mL	N	Y	Y NovoMix 30 FlexPen®	Discard 28 days after 1st use or removal from fridge (< 30°C)
Ryzodeg 70/30® *Formulary One: Inpatient (non-formulary)*	Degludec 70% + Aspart 30% 100units/mL	N	Y	Y Ryzodeg FlexTouch®	Discard 28 days after 1st use or removal from fridge (< 30°C)

Appendix 2. Types of Insulins: Onset, Peak & Duration Times

Type	Brand®	Manufacturer	Nature	Presentation
Mealtime or prandial insulins				
Ultra-short acting (onset in 5–15 minutes, peak at 30-90 minutes, duration for 3-5 hours)				
Faster-acting insulin aspart	Fiasp	Novo Nordisk	Analogue	Clear Solution
Ultra-short acting (onset in 10-15 minutes, peak at 1-1.5 hours, duration for 3-5 hours)				
Insulin lispro	Humalog	Eli Lilly	Analogue	Clear Solution
Insulin aspart	NovoRapid	Novo Nordisk	Analogue	Clear Solution
Insulin glulisine	Apidra	Sanofi	Analogue	Clear Solution
Short acting (onset in 30 minutes, peak at 2-3 hours, duration for 6-8 hours)				
Neutral	Actrapid	Novo Nordisk	Human	Clear Solution
	Humulin R	Eli Lilly	Human	Clear Solution
Basal insulins				
Long acting				
Isophane Onset in 1-2.5 hours Peak at 4-12 hours Duration for 16-24 hours	Humulin NPH	Eli Lilly	Human	Cloudy Solution
	Protaphane	Novo Nordisk	Human	Cloudy Solution
Insulin detemir Onset 1-2 hours Peak at 6-8 hours Duration for 12-24 hours	Levemir	Novo Nordisk	Analogue	Clear Solution
Insulin glargine (100 units/mL) Onset in 1-2 hours No peak Duration for 24-36 hours	Optisulin	Sanofi	Analogue	Clear Solution
Insulin glargine (300 units/mL) Onset in 1-2 hours No peak Duration for 24–36 hours	Toujeo	Sanofi	Analogue	Clear Solution
Premixed insulins				
Short acting with long acting				

Neutral insulin 30% with isophane insulin 70% Onset in 0.5-1 hours Peak at 2-12 hours Duration for 16-24 hours	Humulin 30/70	Eli Lilly	Human	Cloudy Solution
Neutral 30% with isophane 70% Onset in 0.5-1 hours, peak at 2-12 hours, duration for 16-24 hours	Mixtard 30/70	Novo Nordisk	Human	Cloudy Solution
Neutral insulin 50% with isophane insulin 50% Onset in 0.5 hours Peak at 4-8 hours Duration for 24 hours	Mixtard 50/50	Novo Nordisk	Human	Cloudy Solution
Ultra-short acting with long acting				
Insulin lispro 25% with insulin lispro protamine 75% Onset in 10-15 minutes Peak at 1 hour Duration for 16-18 hours	Humalog Mix 25	Eli Lilly	Analogue	Cloudy Solution
Insulin lispro 50% with insulin lispro protamine 50% Onset in 10-15 minutes Peak at 1 hour Duration for 16-18 hours	Humalog Mix 50	Eli Lilly	Analogue	Cloudy Solution
Insulin aspart 30% with insulin aspart protamine 70% Onset in 10-15 minutes Peak at 1 hour Duration for 16-18 hours	NovoMix 30	Novo Nordisk	Analogue	Cloudy Solution
Insulin degludec 70% with insulin aspart 30% Onset in 5–20 minutes Peak at 1 hour Duration for 36–48 hours	Ryzodeg 70/30	Novo Nordisk	Analogue	Clear Solution

NOTES:
Ultra-rapid acting insulin should be administered either up to two minutes before a meal, or at the start of a meal, or up to 20 minutes after starting a meal.

Rapid-acting, pre-mixed and co-formulated insulin should be administered 15 minutes prior to a meal.

Short-acting insulin should be administered 20–30 minutes prior to a meal.

Intermediate- and long-acting basal insulins can be given irrespective of a meal.